Learning-based Speculator (§3)

[inputs]

Expansion-based Token Tree Construction

SSM

Expansion

Expansion

Output 0

Output 1

Output 2

OR

Merge-based Token Tree Construction

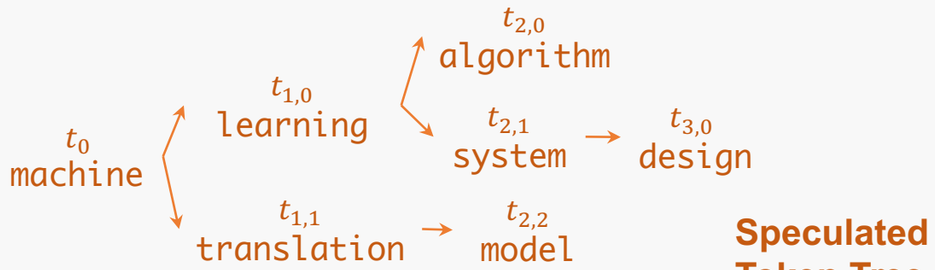
SSM 0 → Output 0

SSM 1 → Output 1

SSM 2 → Output 2

Output 0: [machine] learning algorithm
Output 1: [machine] learning system design
Output 2: [machine] translation model

Sequence Representation of Speculated Tokens



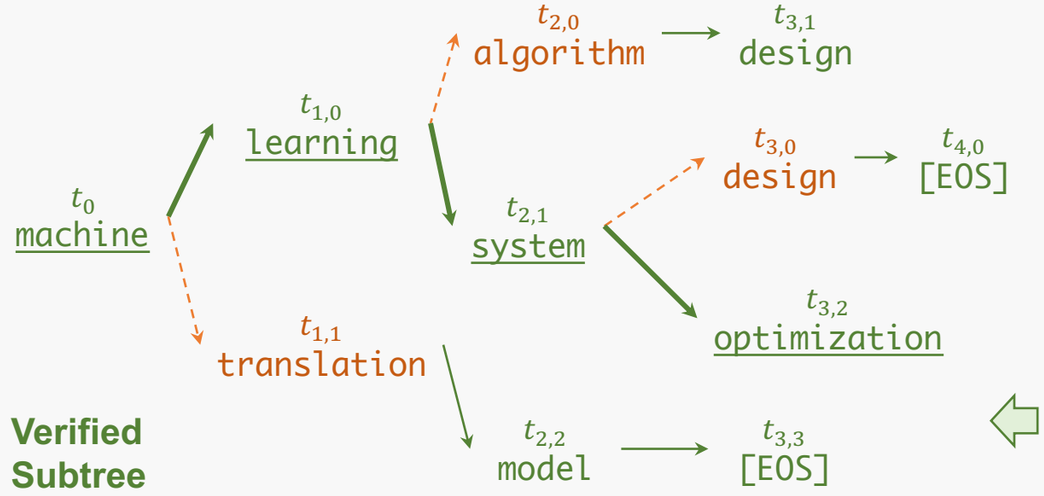
Speculated Token Tree

Verified output: machine learning system optimization

Synchronization

Linearization

Token Tree Verifier (§4)



Verified Subtree

[machine]→learning→algorithm system→design translation→model

Tree-based Parallel Decoding

LLM

learning system design optimization [EOS] model [EOS]